Power in FM

- *Unlike AM,*

$$P_{FM_{Total}} = P_{c_{\text{unmodulated}}}$$

$$P_{FM_{Total}} = P_{c_{\text{modulated}}} + P_{side\ freqs}$$

- $P_{c_{unmodulated}}$ is redistributed between carrier and side frequencies during modulation

Power in FM ..2

- *If V_T is total RMS voltage of unmodulated signal, applied across resistance $R = R_L$*
  - *It has power*

$$P_T = \frac{V_T^2}{R}$$

- *When FM modulation is applied*

$$V_c = J_o V_T$$

  - *Similarly*

$$V_1 = J_1 V_T; \qquad V_2 = J_2 V_T; \qquad V_3 = J_3 V_T; \ etc.$$

  - *Power at carrier frequency will be*

$$P_c = \frac{V_c^2}{R} = \frac{J_0^2 V_T^2}{R}$$

Power in FM ..3

- *The combined power in the 1st set of sidebands*

$$P_{1_T} = 2\frac{V_1^2}{R} = 2\frac{(J_1 V_T)^2}{R}$$

- *The power in the whole signal will be*

$$P_T = P_c + 2P_1 + 2P_2 + 2P_3 + \ldots = \frac{V_T^2}{R}\left[J_o^2 + 2J_1^2 + 2J_2^2 + 2J_3^2 + \ldots\right]$$

Power in FM ..3

Exercise 4

- *An FM signal has a deviation of 4 kHz and a modulating frequency of 2 kHz. Its total power is 7W, developed across a 75 Ω resistor load. The carrier frequency is 180 MHz.*
  - *a) Calculate RMS signal voltage*
  - *b) Calculate RMS voltage at the carrier frequency and each of the first three sets of sidebands*
  - *c) Calculate the frequency of each sideband (for the first three sideband pairs)*
  - *d) Calculate the power at the carrier frequency and at each of the side frequencies found in (c)*

FM

Advantages of FM

Disadvantages of FM

❖ FM is used only at higher frequencies?

FM

- FM is used for?

Phase Modulation

- PM results when
  - Baseband signal varies the phase of the high frequency carrier

- PM signal $\Theta(t) = \omega_c t + K_p v_m(t) = \omega_c t + K_p V_m sin\omega_m t$

- *Phase deviation* $\phi = K_p V_m sin\omega_m t \quad (in\ radians)$

- Peak phase deviation (in rad) $\phi_{max} = \Delta\Theta = K_p V_m = m_p$ *(phase modulation index)*

- *Phase modulation sensitivity* $K_p = {}^{\phi}/_{V_m}$ *rad/volt*

Phase Modulation …

- m_f = FM modulation index

- Given PM, $m_p \; in \; radians$,
  - $m_f = \; m_p$

Exercise 5

- A phase modulator has a sensitivity of $K_p = 4\, rad/V$. How much frequency deviation does it produce with a sine wave input of 3 V peak at a frequency of 1.5 kHz?